University of Cape Town
Department of Computer Science

Computer Science CSC3003S
November Examination, 2019

Read through the entire examination paper before you start writing so that you can answer the easy questions first. The relative importance of a question is indicated by the marks allocated to it. Please write legibly.

This paper consists of **Four Sections:**

Section A: *Advanced Software Design 1*

Section B: *Advanced Software Design 2*

Section C: *Theory of Algorithms*

Section D: *Theory of Computation*

Total Marks: 90	Approximate marks per question are shown in brackets.
Time: 3 hours + 10 minutes reading	The use of calculators is permitted.

Please write your answers in FOUR separate books. Use one book for Advanced Software Design 1, one book for Advanced Software Design 2, and one book each for Theory of Algorithms and Theory of Computation. **Label each of the four books as ASD1, ASD2, ToA and ToC**.

Your attention is drawn to the instructions on your answer book on the front and the back. In particular, "**EVERY CANDIDATE MUST enter the book number and the number of each question answered (in the order in which it has been answered)**" on the table provided on the front cover of the first book.

Please **do this for the front cover of the different books used for each section.**

This is needed so that we can allocate marks and not miss questions. **A list of sub-questions is *not* required.**

Section A. [20]

Question 1. **[12]**

a) Name 3 benefits of iterative development. [3]

b) What are the purposes of the Daily Scrum meeting? [2]

c) Extreme programming maintains that changes cannot be reliably anticipated. Why, then, does it advocate for refactoring? [1]

d) We introduced the 3M's of dealing with risk: Mitigation, Monitoring and Management. Briefly explain what is meant by each of these "M"s. [6]

Question 2. **[8]**

a) Contrast the iterative approach to software development with the waterfall approach, in terms of the following: timescale, requirements, and product release(s). [6]

b) Which is typically less risky, the waterfall approach or an iterative approach? Why? [2]

Section B. [22]

Question 3. **[7]**

a) What is meant by software architecture? [2]

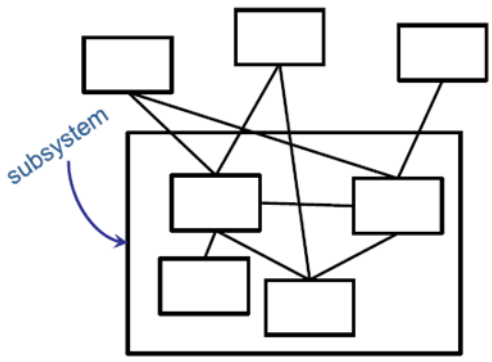
b) What is the advantage of defining a software component as a particular type? [1]

c) Indicate two benefits of design patterns? [2]

d) What are the potential disadvantages of a layered architecture? [2]

Question 4. **[6]**

The first draft of the design of a new system results in the following architectural design.



a) What is the obvious problem with this design and why is it undesirable? [2]

b) Name the Gang of Four design pattern best suited to resolving the problem with the architectural design above, explaining what is does. Also redraw the architectural design, implementing the design pattern to illustrate how it is used. [4]

Question 5. **[9]**

a) What does the acronym CI/CD stand for in the context of Software Engineering? [2]

b) What are some of the challenges with native and web apps, which Progressive Web Apps try and resolve? [3]

c) What is a blockchain? [2]

d) What are the properties of a good microservice? [2]

Section C. [24]

Question 6. **[2]**

Which two statements below are True?

a) $O(n + \log n) = O(n)$

b) $\Theta(\log^2 n) = \Theta(\log n)$

c) if $x \in O(n \log n)$ then $x \in \Omega(n)$

d) if $x \in \Theta(2^n)$ then $x \in \Omega(n)$

Question 7. **[2]**

How would the product of 2 integers ***m*** and ***n*** be computed using each technique below? You need just 1 line per answer.

a) Brute force

b) Decrease by a constant

c) Decrease by a constant factor

d) Divide and conquer

Question 8. **[2]**

Given an MxN map, where you start at position (1,1), the top left corner, you want to know how many routes there are to get to (M,N), the bottom right corner of the map. The only moves you can make are to go down or to the right, and there are some grid positions – i.e. some cells (R,C) – that you cannot move into because of an obstacle there. For example, if the map is 4x4 and there is only 1 obstacle in position (2,3), then the number of routes is 11.

Give an algorithm to solve this problem. Remember, any point on the map can only be reached by coming either from the left or else from above.

Question 9. **[4]**

Strassen's algorithm for matrix multiplication uses the following formula:

$$\begin{bmatrix} a00 & a01 \\ a10 & a11 \end{bmatrix} * \begin{bmatrix} b00 & b01 \\ b10 & b11 \end{bmatrix} = \begin{bmatrix} m1 + m4 - m5 + m7 & m3 + m5 \\ m2 + m4 & m1 + m3 - m2 + m6 \end{bmatrix}$$

where

m1 = (a00 + a11) * (b00 + b11); m2 = (a10 + a11) * b00;

m3= a00 * (b01 – b11); m4 = a11 * (ba0 – b00);

m5 = (a00+a01) * b11; m6 = (a10 – a00) * (b00 + b01);

m7 = (a01 – a11) * (b10 + b11)

So, e.g., to multiply two 2 by 2 matrices requires 7 multiplications (and 18 additions / subtractions), compared to 8 multiplications (and 4 additions) without this method.

a) Suppose you want to use Strassen's algorithm to multiply two 15 by 15 matrices. What problem do you face and how would you solve it?

b) What would the values of ***a, b*** and ***d*** be if we used the Master Theorem to find the efficiency of Strassen's algorithm in terms of the number of multiplications? **Just give the 3 values, showing clearly which is which**.

Master Theorem: If $T(n) = aT(n/b) + f(n)$ then $T(n)$ $O(n^d)$ if $a < b^d$

$T(n)$ $O(n^d \log n)$ if $a = b^d$

$T(n)$ $O(n^{\log_b a})$ if $a > b^d$

Question 10. **[2]**

Outline how you could **most efficiently** solve the problem of finding the K'th largest value in a huge, unsorted/unordered array of numbers. Give 1 to 3 lines to show the general idea rather than details.

Question 11. **[6]**

A collection of gems is being sold. To prevent a single buyer from purchasing them all, the cost increases greater than linearly with the number of gems bought. The total cost to buy K gems is f(K), where f(K) is the sum of j * (K/j) for all integers $0 < j < K$. Note that K/j here

uses integer division, so the result has no fractional part. For example: 8/3 would equal 2. As an example, to buy 5 gems, the cost, f(5), would be:

1 * (5/1) + 2 * (5/2) + 3 * (5/3) + 4 * (5/4) = 1*5 + 2*2 + 3*1 + 4*1 = 16

You have to find the largest number of gems that can be bought for a given amount of money. That is, given an amount of money, N, find the greatest value of K for which f(K) ≤ N. Note that the function f(K) is strictly increasing, and that K is guaranteed to be at most 1 000 000.

a) Outline a brute force algorithm (in English or pseudocode) to solve this problem. [2]

b) Now give pseudocode for a more efficient algorithm to solve this problem. [3]

c) Give the efficiency of each of the two algorithms above. [1]

Question 12. **[3]**

Show by means of a **small** worked example of your own choice how to generate all 2^n subsets of a given set of n elements.

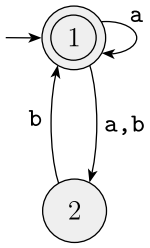
Question 13. **[3]**

Show by means of a small worked example of your own choice how Warshall's algorithm for finding the transitive closure of a graph works. Just show the initialised array, and then the array obtained after the first iteration of the algorithm. This is enough to show how it works. Your example should have about 4 nodes and about 4 edges.

Section D. [24]

Question 14. **[4]**

Construct a DFA that accepts the same language as the following NFA:



Question 15. **[3]**

Suppose we want to prove that for any regular language A, its complement is also regular. We know that if A is regular, then there is a Non-Deterministic Finite Automaton N such that N accepts A. From N we now construct a new Non-Deterministic Automaton N' by swapping the accept states and the reject states in N. That is, every accept state in N is a reject state in N' and every reject state in N is an accept state in N. Does N' accept the complement of A? If it does, explain why in detail. If it does not, explain why not, indicate what language it accepts.

Question 16. **[4]**

Explain how you would prove that every regular expression R can be used to construct an NFA that accepts exactly the language generated by R.

Question 17. **[3]**

Prove that the following language is not regular: $L = \{w \mid w \in \{a,b,c,d,e\}^* \text{ and } w = w^r\}$. Note that L consists of the palindromes made up of the letters a, b, c, d, and e.

Question 18. **[4]**

Convert the following CFG to Chomsky normal form:

$S \rightarrow ASA \mid aB$

$A \rightarrow B \mid \varepsilon$

$B \rightarrow b$

Question 19. **[3]**

Given a context-free language L and a PDA P that accepts it, explain how to construct a PDA that accepts L*.

Question 20. **[3]**

If a language is Turing-decidable, is it also Turing-recognisable? If it is, prove it. If not, give an example of a language that is Turing-recognisable but not Turing-decidable.

-END-